A neutron star orbiting Saturn? Impossible! A black hole accompanying the neutron star, with both orbiting Saturn? Even more impossible! Saturn would long ago have been destroyed, and the star's and hole's gravity would long ago have disrupted the orbits of all the Sun's planets, including Earth. With disrupted orbit, the Earth would have been carried close to the Sun and then far away. We would have been fried, frozen, and killed.

But there the waves were. Unequivocally emerging from near Saturn.

Professor Brand could find only one explanation: The waves must emerge from a wormhole that orbits Saturn. And their source, the black hole and neutron star, must be on the other end of the wormhole (Figure 16.5). The waves traveled outward from the star and hole. Small portions of the waves were captured by the wormhole, traveled through it, and then spread outward through the solar system with a small portion reaching Earth and passing through the LIGO gravitational wave detector.

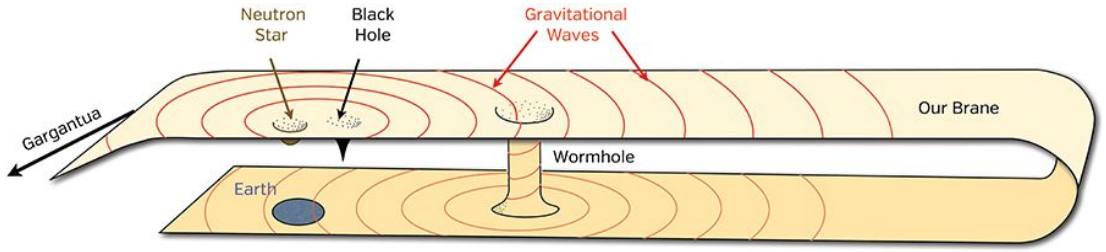


Fig. 16.5. Gravitational waves travel through the wormhole to Earth.

Origin of This Story

A brief variant of this story was in the original 2006 treatment for *Interstellar* that Lynda Obst and I wrote. However, gravitational waves did not play a significant role in the rest of our treatment, nor in the subsequent screenplay that Jonathan Nolan wrote and Chris rewrote.